Delhi Public School

Subject: Chemistry | Grade: 11th | Class: 11th

Time: 2 Hours | Maximum Marks: 30

Name: ________________________ Roll No: ___________ Section: ___________

General Instructions:

1. *Read all questions carefully before answering.*
2. *Write neatly and legibly.*

Section A (Objective Type Questions)

Answer all questions. For multiple-choice questions, select the single best option. For true/false questions, state your choice and provide a brief justification.

Q1. Which of the following properties of cathode rays is independent of the nature of the gas present in the discharge tube? [Easy] **[1]**

a. Color of fluorescence produced on the walls
b. Charge-to-mass (e/m) ratio of the constituent particles
c. Deflection behavior in an external magnetic field
d. None of the above

Q2. According to Bohr's theory, the angular momentum of an electron in the 4th orbit of a hydrogen atom is: [Moderate] **[1]**

a. h/π
b. 2h/π
c. 4h/π
d. h/2π

Q3. What is the maximum number of electrons that can be accommodated in a subshell for which the azimuthal quantum number l = 3? [Easy] **[1]**

a. 6
b. 10
c. 14
d. 18

Q4. The electronic configuration of Chromium (Z = 24) in its ground state is [Ar] 3d^5 4s^1 instead of [Ar] 3d^4 4s^2. This variation is best explained by which of the following? [Moderate] **[1]**

a. The Aufbau Principle
b. Pauli's Exclusion Principle
c. The extra stability associated with half-filled d-subshells
d. Heisenberg's Uncertainty Principle

Q5. What is the de Broglie wavelength associated with a macroscopic particle of mass 1.0 x 10^-6 kg moving with a velocity of 10 m/s? (Planck's constant, h = 6.626 x 10^-34 J s) [Hard] **[1]**

a. 6.626 x 10^-29 m
b. 6.626 x 10^-31 m
c. 6.626 x 10^-39 m
d. 6.626 x 10^-25 m

Q6. State whether the following statement is True or False, and provide a brief justification: 'An orbital can accommodate up to two electrons, provided they have the same spin quantum number (ms).' [Moderate] **[3]**

Section B (Short Answer Questions)

Answer all questions in brief, showing relevant steps or reasoning where applicable.

Q7. State two major drawbacks of Rutherford's nuclear model of the atom. [Easy] **[2]**

Q8. Using Hund's Rule of Maximum Multiplicity, write the ground-state electronic configuration of a Nitrogen atom (Z = 7) and explain the spatial distribution/spin of its 2p electrons. [Moderate] **[2]**

Q9. An electron resides in a 4d subshell. State the value of its principal quantum number (n), azimuthal quantum number (l), and list all possible values of its magnetic quantum number (ml). [Moderate] **[2]**

Section C (Long Answer Questions)

Answer all questions. Provide detailed explanations, calculations, or justifications as required.

Q10. Compare the Bohr model of the atom with the Quantum Mechanical model. Highlight three major conceptual differences, specifically emphasizing the ideas of orbits vs. orbitals and how Heisenberg's Uncertainty Principle is addressed. [Hard] **[4]**

Q11. State the Aufbau Principle. Explain how the (n + l) rule determines the relative energy levels of orbitals, and use this rule to demonstrate why the 4s orbital is filled before the 3d orbital in transition elements. [Moderate] **[4]**

Q12. Calculate the wavelength (in meters) of the spectral line emitted when an electron in a hydrogen atom undergoes a transition from the energy level n = 4 to the energy level n = 2. (Rydberg constant, RH = 1.09677 x 10^7 m^-1). Identify the spectral series to which this transition belongs. [Hard] **[4]**

Q13. Analyze the following sets of quantum numbers (n, l, ml, ms). Identify which of these sets are physically impossible for an electron within an atom, and provide a clear reason for each invalid set: a) (2, 2, 1, +1/2) b) (3, 1, 0, -1/2) c) (1, 0, 0, +1/2) d) (2, 1, -2, -1/2) [Moderate] **[4]**

— End of Question Paper —